ShenYi (Tommy) Wang

Toronto, Ontario | github.com/Grimshock1015
437-353-9100 | www.linkedin.com/in/tommy-wang1015 | shenyi.wang@mail.utoronto.ca

SKILLS:

Languages & Web: Python, C, Java, SQL, JavaScript, TypeScript, HTML/CSS, PHP, WebRTC
AI & ML: PyTorch, scikit-learn, XGBoost, LangChain, RAGAS
Databases: MySQL, Firebase, MongoDB, Milvus, NeDB
Tools & Practices: Git, Linux, Docker, Android Studio; Agile / Scrum

EDUCATION:

University of Toronto 2023 - 2027
Honours BSc, Computer Engineering Specialist Program | University of Toronto Scholar
National University of Singapore | Exchange Student 2026

WORK / INTERNSHIP EXPERIENCE:

Software Design Intern – GienTech China | *Python, LangChain, Milvus* Summer 2025
- Built a medical-device Q&A prototype with Python, LangChain, Xinference, and Milvus, using retrieval-augmented generation (RAG) to answer product questions from company manuals.
- Converted medical-device manual content into searchable embeddings and retrieved relevant passages from Milvus, supplying source context for the prototype's product answers.
- Assessed prototype response quality with RAGAS on the company's 500-question benchmark; the pipeline achieved a reported internal benchmark score of approximately 90%.
- Delivered the prototype with a 3-member cross-functional Agile team, using daily standups, iterative debugging, and code reviews to integrate and validate the Q&A workflow.

PROJECT EXPERIENCE:

Apache Superset – Google Sheets Export Proposal Oct 2026 – Present
- Contributed to a technical proposal to replace manual CSV/XLSX uploads with direct Google Sheets export, extending the existing SQL Lab design to charts, dashboards, and drill results.
- Specified synchronous chart export and Celery-based dashboard jobs with status polling, producing an execution design intended to avoid request timeouts during multi-chart exports.
- Documented extension-menu integration and one spreadsheet tab per chart, including export metadata, so the proposed workflow preserves chart context when results are shared.

Daily Emission Tracker – Planetze | *Java, Android, Firebase* Dec 2024
- Developed Java features for habit adoption, logging, and recommendation in an external company's Android carbon-tracking course project, enabling users to track daily sustainability habits.
- Implemented per-user habit histories through Firebase, duplicate-adoption checks, and date-based logging, enabling users to save adopted habits and review completion in calendar views.
- Built rule-based recommendations from users' most frequent activity categories and added keyword, category, and impact filters, allowing users to discover habits relevant to their activity patterns.
- Delivered habit-management features as part of the completed Android app; earned a 10/10 individual score, while the team received 96% and selection for an external-company showcase.

Global Game Jam | *Godot, GDScript, JSON* Jan 2026
- Developed a playable 2D narrative puzzle game in Godot during a 48-hour Global Game Jam, combining scripted interactions, UI, and branching dialogue into the team's prototype.
- Implemented Area2D triggers, scripted interactions, UI elements, and puzzle logic in GDScript, connecting player actions to in-game events and puzzle progression.
- Created JSON-based branching dialogue to represent player choices and narrative paths, enabling the game to display different story responses for different selections.

Toronto Safety Index Prediction | *Python, scikit-learn, XGBoost* Dec 2025
- Built a 0–1 safety-index target for 158 Toronto neighbourhoods by mapping police-reported crime categories to severity weights, turning monthly incident records into a regression dataset.
- Engineered 1–3-month lag features and trained linear, ridge, random-forest, and XGBoost regressors on 2020–2023 data, using 2024–2025 as a chronological holdout for model comparison.
- Tuned XGBoost tree depth and estimator count with 5-fold TimeSeriesSplit, selecting hyperparameters through chronological cross-validation before evaluating the model on the held-out period.
- Achieved the lowest reported test MSE among four models: XGBoost reached 0.00270 MSE and 0.8176 R^2, approximately 3.6% lower MSE than the linear baseline (0.00280).